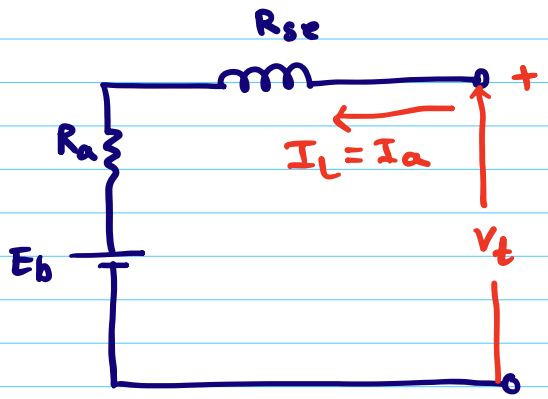
EE2005 : TUTORIAL - 5 on DC Machines-II

1. A 200 V DC series motor runs at 700 rpm when operating at its full load current of 20 A. The motor resistance is 0.5 Ω and the magnetic circuit can be assumed unsaturated. What will be the speed if (i) the load torque is increased by 44% (ii) the motor current is 10 A.



$V_t = 200\,V$ $\quad R_s = 0.5\,\Omega$

where $R_s = R_a + R_{se}$

$I_{a(full)} = 20\,A$

$N_{(full)} = 700\ RPM$

$T = K_t I_a^2$ & $E_b = K_t I_a \omega$

we write

$$E_b = V_t - I_a R_s$$

$$K_t = \frac{E_b}{\omega I_a} = \frac{200 - 20 \times 0.5}{\frac{2\pi * 700}{60} \times 20}$$

$$k_t = 0.12959 \text{ V/rad/sec}$$

Load torque

$$T_1 = k_t * I_{a1}^2$$

$$T_1 = 0.12959 \times 20^2$$

$$T_1 = 51.8390 \text{ N-m}$$

(i) given

$$T_2 = 1.44 * T_1$$

$$T_2 = 74.6482 \text{ N-m}$$

But k_t is not changed so the proportionality holds

i.e., $T \propto I_a^2$

$$\frac{T_1}{T_2} = \left(\frac{I_{a1}}{I_{a2}}\right)^2$$

$$\rightarrow \quad I_{a2} = I_{a1}\sqrt{\frac{T_2}{T_1}}$$

$$= 20 \times \sqrt{\frac{1.44}{1}}$$

$$I_{a2} = 24 \text{ A}$$

$$E_{b_2} = V_t - I_{a_2} \times R_s$$
$$= 200 - 24 \times 0.5$$
$$= 188\ V$$

$$\boxed{E_{b_2} = 188\ V}$$

$$K_f \times I_{a_2} \times \omega_2 = 188\ V$$

$$\omega_2 = \frac{188}{24 \times 0.12959}$$

$$\boxed{\omega_2 = 60.4435\ \text{rad/sec}}$$

$$N_2 = \frac{60}{2\pi}\,\omega_2$$

$$\boxed{N_2 = 577.1929\ \text{RPM}}$$

(ii)

Given $\boxed{I_{a_3} = 10\ A}$

$$E_{b3} = V_t - I_{a3} R_s$$

$$= 200 - 10 \times 0.5$$

$$E_{b3} = 195\ V$$

$$\Rightarrow \quad K_t I_{a_3} \omega_3 = 195$$

$$\omega_3 = \frac{195}{K_t I_{a_3}}$$

$$= \frac{195}{0.12959 \times 10}$$

$$\omega_3 = 150.4658 \text{ rad/sec}$$

$$N_3 = \frac{60}{2\pi} \omega_3$$

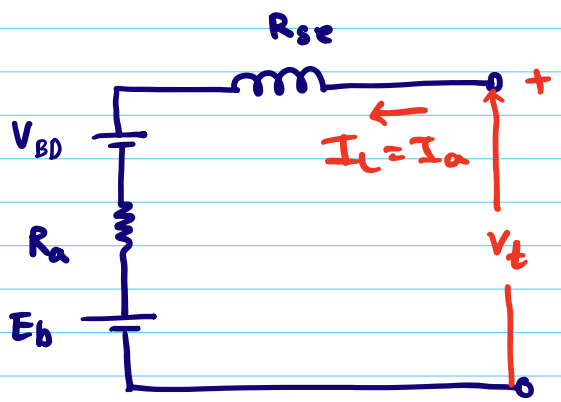
$$N_3 = 1436.8421 \text{ RPM}$$

2. A 220 V DC series motor draws full load line current of 38 A at the rated speed of 600 rpm. The motor has armature resistance of 0.4 Ω and the series field resistance is 0.2 Ω. The brush voltage drop irrespective of load is 3.0 V, find :

   (a) The speed of the motor when the load current drops to 19 A.

   (b) The speed on removal of load when the motor takes only 1 A from supply.

   (c) The internal horse power developed in each of the above cases. Neglect the effect of armature reaction and saturation.



V_{BD} : Brush Voltage Drop

$V_t = 220\ V$ $\qquad$ $R_a = 0.4\ \Omega$

$V_{BD} = 3\ V$ $\qquad$ $R_{se} = 0.2\ \Omega$

$I_{a(full)} = 38\ A$

$N_{(full)} = 600\ RPM$

$$E_b = V_t - I_a (R_a + R_{se}) - V_{BD}$$

$$E_{b_1} = 220 - 38\ (0.4 + 0.2) - 3$$

$$E_{b_1} = 194.2\ V$$

$$\Rightarrow \quad E_b = k_t I_a \omega$$

$$E_{b1} = k_t I_{a1} \omega_1$$

$$k_t = \frac{E_{b1}}{I_{a1}\ \omega_1}$$

$$= \frac{194.2}{38\ \frac{2\pi \times 600}{60}}$$

$$k_t = 0.081337 \text{ V/rad/sec}$$

(i) $I_{a2} = 19$ A

$$E_{b2} = V_t - I_{a2}(R_a + R_{se}) - V_{BD}$$

$$= 220 - 19 * (0.4 + 0.2) - 3$$

$$E_{b2} = 205.6 \text{ V}$$

$$k_t I_{a2} \omega_2 = 205.6$$

$$\omega_2 = \frac{205.1}{k_t I_{a2}}$$

$$= \frac{205.6}{0.081336 \times 19}$$

$$\omega_2 = 133.0405 \text{ rads/sec}$$

$$N_2 = 1270.4428 \text{ RPM}$$

(or) Use the relation

$E_b \propto K_e \omega \propto K_a \phi \omega \propto I_a \omega$ ($\because \phi \propto I_a$ in series machine)

$$\frac{E_{b2}}{E_{b1}} = \frac{I_{a2}}{I_{a1}} \times \frac{\omega_2}{\omega_1}$$

$$\omega_2 = \frac{E_{b2}}{E_{b1}} \times \frac{I_{a1}}{I_{a2}} \times \omega_1$$

$$\omega_2 = \frac{205.6}{194.2} \times \frac{38}{19} \times \frac{2\pi 600}{60}$$

$$\omega_2 = 133.04 \text{ rads/sec}$$

$$N_2 = 1270.4428 \text{ RPM}$$

(ii)

$$I_{a3} = 1\ A$$

$$E_{b3} = V_t - I_{a3} \times (R_a + R_{sc}) - V_{BD}$$

$$= 220 - 1 \times (0.4 + 0.2) - 3$$

$$= 216.4\ V$$

$$E_{b3} = K_t I_{a3} \omega_3$$

$$\omega_3 = \frac{E_{b3}}{K_t I_{a3}}$$

$$= \frac{216.4}{0.0813365 \times 1}$$

$$\omega_3 = 2660.5504\ \text{rad/sec}$$

$$N_3 = \frac{60}{2\pi} \omega_3$$

$$N_3 = 25406.38517\ \text{RPM}$$

(OR) Use the relation

$E_b \propto K_e \omega \propto K_a \phi \omega \propto I_a \omega$ ($\because \phi \propto I_a$ in Series machine)

$$\frac{E_{b3}}{E_{b1}} = \frac{I_{a3}}{I_{a1}} \times \frac{\omega_3}{\omega_1}$$

$$\omega_3 = \frac{E_{b2}}{E_{b1}} \times \frac{I_{a1}}{I_{a2}} \times \omega_1$$

$$\omega_3 = \frac{216.4}{194.2} \times \frac{38}{1} \times \frac{2\pi 600}{60}$$

$$\omega_3 = 2660.5504\ \text{rad/sec}$$

$$N_3 = 25406.38517$$

(iii) Mechanical power developed,

$$P = E_b I_a \quad \text{(in watts)}$$

$$= \frac{E_b I_a}{746} \quad \text{(in h.p.)}$$

(i) For $I_{a2} = 19\,A$

$$P_2 = \frac{E_{b2} I_{a2}}{746}$$

$$= \frac{205.6 \times 19}{746}$$

$$P_2 = 5.2385 \text{ hp}$$

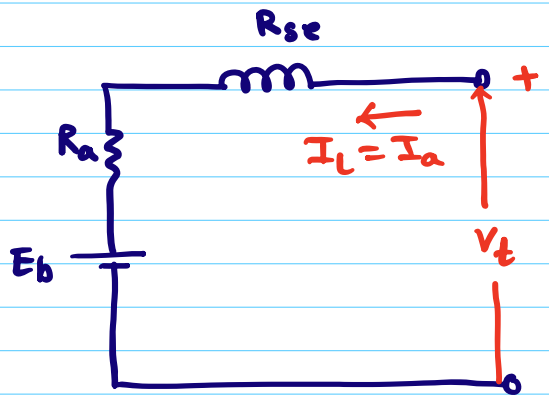
(ii) For $I_{a3} = 1\,A$

$$P_3 = \frac{E_{b3} I_{a3}}{746}$$

$$= \frac{216.4 \times 1}{746}$$

$$P_3 = 0.290197 \text{ hp}$$

3. A 440 V series motor has a total series resistance (including field) of 0.8 Ω. It takes 110 A to give the rated output at 800 rpm. Determine the resistance to be added to obtain rated torque (i) At starting (ii) At 600 rpm. Assume that the flux is proportional to the current.



$V_t = 440\,V$

$R_s = 0.8\,\Omega = (R_a + R_{se})$

$I_a = 110\,A \longleftarrow N_{(rated)} = 800\ RPM$

$E_b = V_t - I_a R_s$

$= 440 - 110 * 0.8$

$E_b = 352\,V$

also

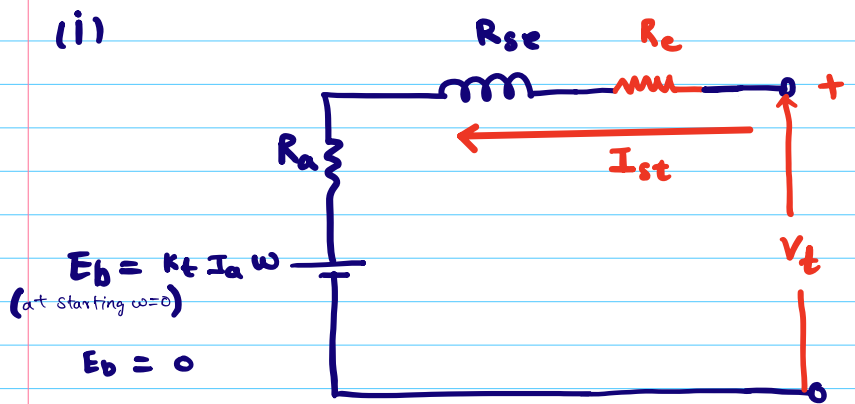
$E_b = k_t I_a \omega$

$k_t = \dfrac{E_b}{I_a \times \omega}$

$$K_t = \frac{312}{110 \times \frac{2\pi \times 800}{60}}$$

$$K_t = 0.0581971 \ \text{V/rad/sec}$$

(i)



To obtain the rated torque at starting

so $T_{st} \propto I_{st}^2$

$$T_{st} = K_t I_{st}^2$$

If T_{st} = Rated torque

then $I_{st} = I_{rated} = 110A$

$$\Rightarrow \quad T_{st} = 0.38856 \times 110^2$$

$$T_{st} = 462.18595 \ \text{N-m}$$

so KVL

$$V_t = \cancel{E_b}^{\,0} + I_a(R_s + R_c)$$

$$V_t = I_a(R_s + R_c)$$

$$R_s + R_c = \frac{V_t}{I_a}$$

$$R_c = \frac{V_t}{I_a} - R_s$$

$$= \frac{440}{110} - 0.8$$

$$R_c = 3.2\ \Omega$$

(ii)

$$N = 600\ \text{RPM}$$

$$\omega = \frac{2\pi}{60} N$$

$$\omega = 62.8319\ \text{rad/sec}$$

$$V_t = E_b + I_a (R_s + R_c)$$

To obtain 600 RPM with rated current 110A

$$V_t = k_t I_a \omega + I_a (R_s + R_c)$$

$$R_c = \frac{V_t}{I_a} - k_t \omega - R_s$$

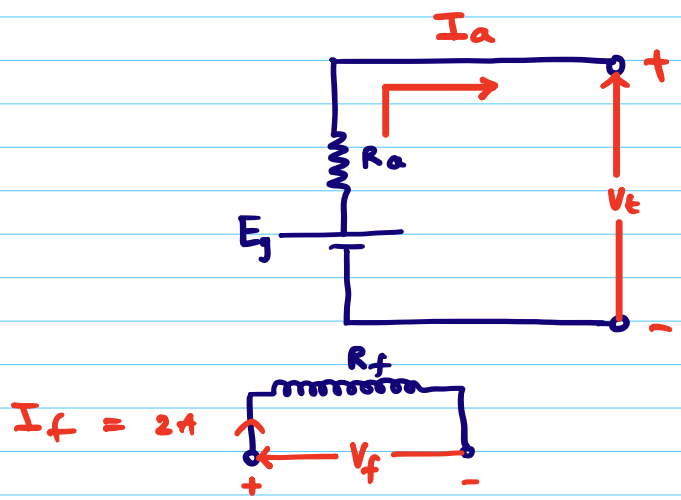
$$= \frac{440}{110} - 0.038197 \times 62.8319 - 0.8$$

$$R_c = 0.8\ \Omega$$

4. A separately excited DC generator has no-load voltage of 120 V at a field current of 2 A, when driven at 1500 rpm. Assuming that it is operating on the straight line portion of its saturation curve, calculate :

   (a) The generated voltage when the field current is increased to 2.5 A.

   (b) The generated voltage when the speed is reduced to 1400 rpm and the field current is increased to 2.84 A.



At no-load

$$E_{go} = V_t = 120\ V$$

$$E_{go} = k_c \omega = 120$$

$$k_c = \frac{120}{\omega}$$

$$= \frac{120}{\frac{2\pi \times 1500}{60}}$$

$$k_c = 0.7639 \quad V/rad/sec$$

(a) $E_g \propto K_c \omega \propto K_c \phi \omega \propto I_f K' \omega$

$$\frac{E_{g_0}}{E_{g_1}} = \frac{I_{f_0}}{I_{f_1}}$$

Assumed that speed is constant. i.e., 1500 RPM.

$$E_{g_1} = E_{g_0} \frac{I_{f_1}}{I_{f_0}}$$

$$= 120 \frac{2.5}{2}$$

$$E_{g_1} = 150\ V$$

(b) $N_2 = 1400$ RPM

$I_{f_2} = 2.84$ A

$E_g \propto I_f N$

$$\frac{E_{g_0}}{E_{g_2}} = \frac{I_{f_0}}{I_{f_2}} \times \frac{N_0}{N_2}$$

$$E_{g_2} = \left(\frac{I_{f_2}}{I_{f_0}}\right)\left(\frac{N_2}{N_0}\right) E_{g_0}$$

$$= \frac{2.84}{2} \times \frac{1400}{1500} \times 120$$

$$E_{g_2} = 159.04\ V$$

(OR)

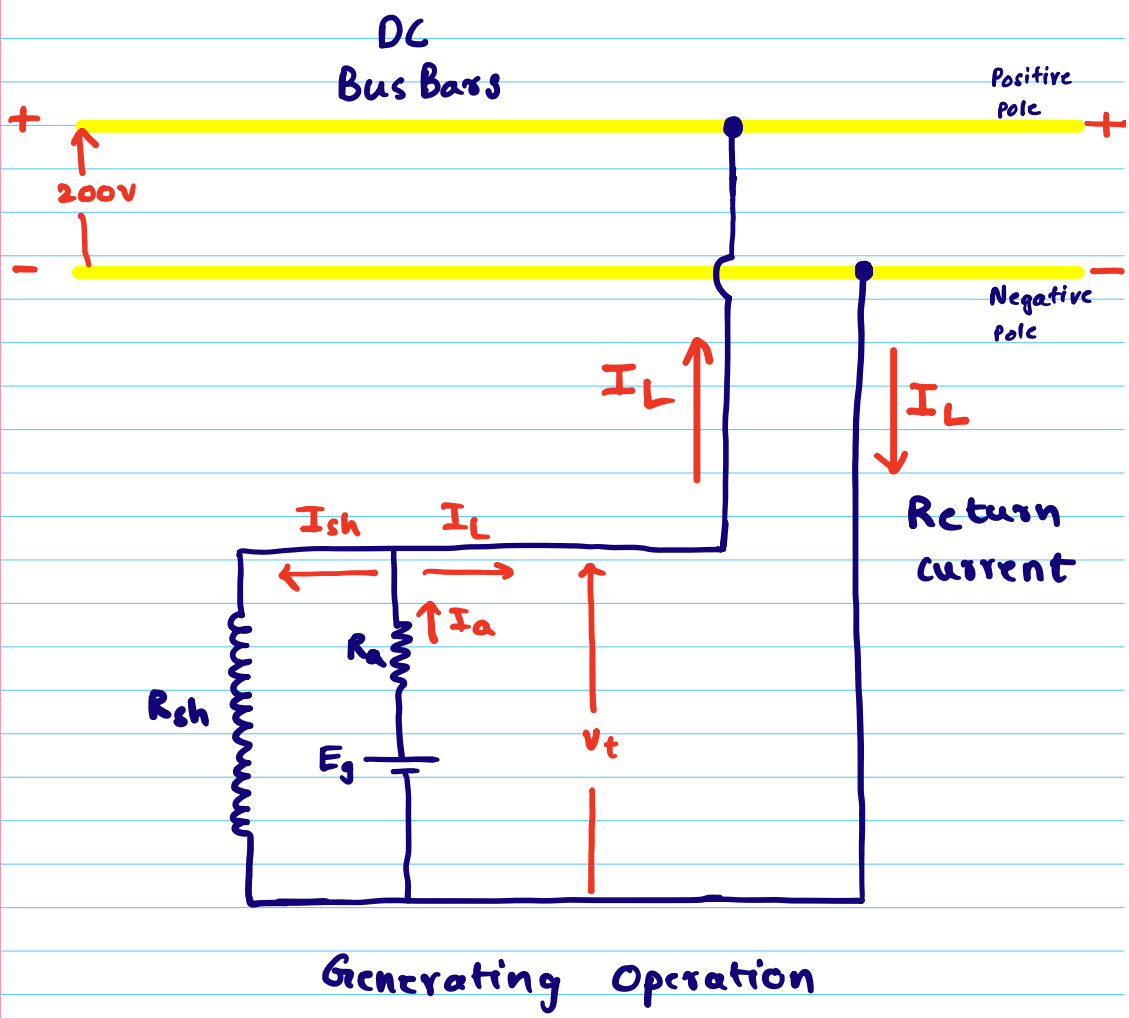
$$\frac{E_{g_1}}{E_{g_2}} = \frac{I_{f_1}}{I_{f_2}} \times \frac{N_1}{N_2}$$

$$E_{g_2} = \left(\frac{I_{f_2}}{I_{f_1}}\right)\left(\frac{N_2}{N_1}\right) E_{g_1}$$

$$= \frac{2.84}{2.5} \times \frac{1400}{1500} \times 150$$

$$E_{g_2} = 159.04\ V$$

5. A DC 60 kW shunt wound generator running at 500 rpm is supplying full load to a bus bar at 200 V. If the prime-mover input to the generator is removed and the machine now operates taking 5 kW from the bus bar, at what speed will it operate? The armature and field resistances are 0.1 Ω and 100 Ω respectively. Brush contact drop may be taken as 2 V. Neglect armature reaction.



$R_a = 0.1\Omega$

$R_{sh} = 100\Omega$ $\quad$ B.D = 2 V

$V_t = 200V$ $\quad$ $P_o = 60KW$ $\quad$ N = 500 RPM

$$E_g = V_t + I_a R_a + B.D$$

Given $\quad P_o = 60KW = V_t I_L$

$$60KW = 200 \times I_L$$

$$I_L = \frac{60 \times 10^3}{200}$$

$$I_L = 300\,A$$

Now $\quad I_a = ?$

By KCL $\quad I_a = I_{sh} + I_L$

so $\quad I_{sh} = \frac{V_t}{R_{sh}}$

$$= \frac{200}{100}$$

$$I_{sh} = 2\,A$$

$$I_a = I_{sh} + I_L$$

$$= 2 + 300$$

$$I_a = 302 \text{ A}$$

Now

$$E_g = V_t + I_a R_a + B.D$$

$$= 200 + 302 * 0.1 + 2$$

$$E_g = 232.2 \text{ V}$$

to find K_c

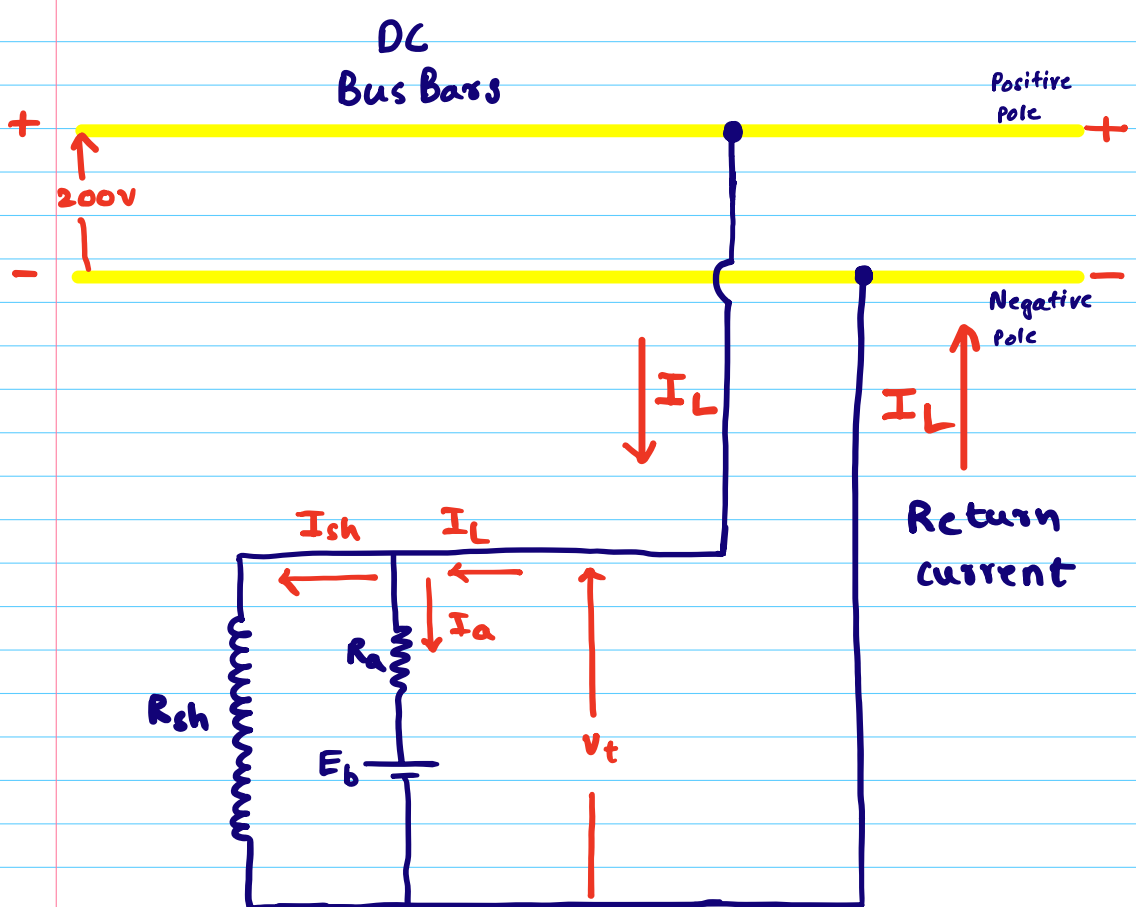
$$E_g = K_c \omega$$

$$K_c = \frac{E_g}{\omega}$$

$$= \frac{E_g}{\frac{2\pi}{60} \times N}$$

$$= \frac{232.2}{\frac{2\pi}{60} \times 500}$$

$$K_c = 4.435 \text{ V/rad/sec}$$



Motoring Operation

Given $P_i = 5KW$

$$I_L = \frac{5000}{200}$$

$$I_L = 25A$$

Already

$$I_{sh} = 2\ A$$

so $$I_a = I_L - I_{sh}$$

$$= 25 - 2$$

$$I_a = 23\ A$$

Apply KVL

$$V_t = E_b + I_a R_a + B.D$$

$$E_b = V_t - B.D - I_a R_a$$

$$= 200 - 2 - 23 \times 0.1$$

$$E_b = 195.7\ V$$

we know

$$E_b = K_e \omega$$

$$\omega = \frac{E_b}{K_e}$$

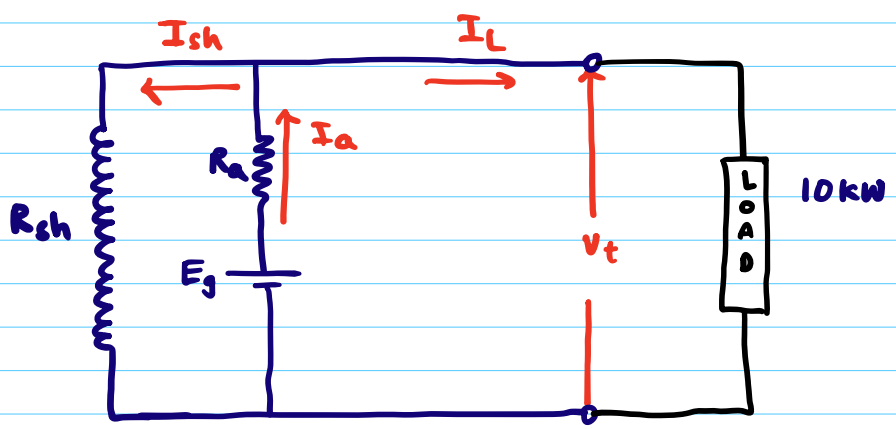
$$= \frac{195.7}{4.4347}$$

$$\omega = 44.1293\ \text{rad/sec}$$

$$N = \frac{60}{2\pi}\,\omega$$

$$N = 421.40296 \text{ RPM}$$

6. A DC shunt generator with an armature resistance of 0.75 Ω and a field resistance of 125 Ω supplies a load of 10 kW at a terminal voltage of 250 V. Determine the efficiency of the generator assuming the iron losses, friction losses and windage losses amount to 600 W.



Generating Operation

Given $V_t = 250V$

$P_o = 10\,KW$

$P_{loss} = 600\,W$

$R_a = 0.75\,\Omega$

$R_{sh} = 125\,\Omega$

$$P_o = V_t \times I_L$$

$$I_L = \frac{P_o}{V_t}$$

$$= \frac{10K}{250}$$

$$I_L = 40\,A$$

$$I_{sh} = \frac{V_t}{R_{sh}}$$

$$= \frac{250}{125}$$

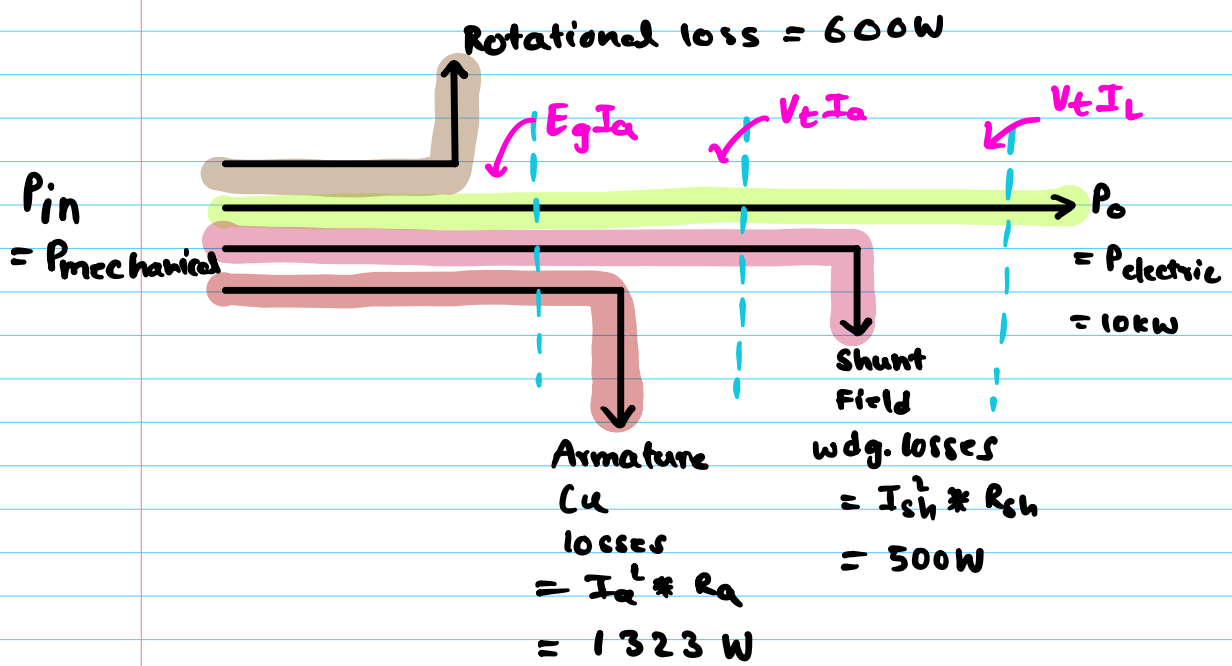
$$I_{sh} = 2\,A$$

By KCL,

$$I_a = I_L + I_{sh}$$

$$= 40 + 2$$

$$I_a = 42\,A$$



$$P_{input} = P_o + P_{Fixed} + I_a^2 R_a + I_{sh}^2 R_{sh}$$

$$= 10 \times 10^3 + 600 + 42^2 * 0.75 + 2^2 \times 125$$

$$= 12423\ W$$

$$\% \text{ Efficiency} = \%\eta = \frac{P_{output}}{P_{input}} * 100$$

$$= \frac{10 \times 10^3}{12423} * 100$$

$$\%\eta = 80.4958\ \%$$

7. A separately excited dc generator is rated 6 kW, 120 V, 1200 rpm, and has the following open circuit characteristics at rated speed.

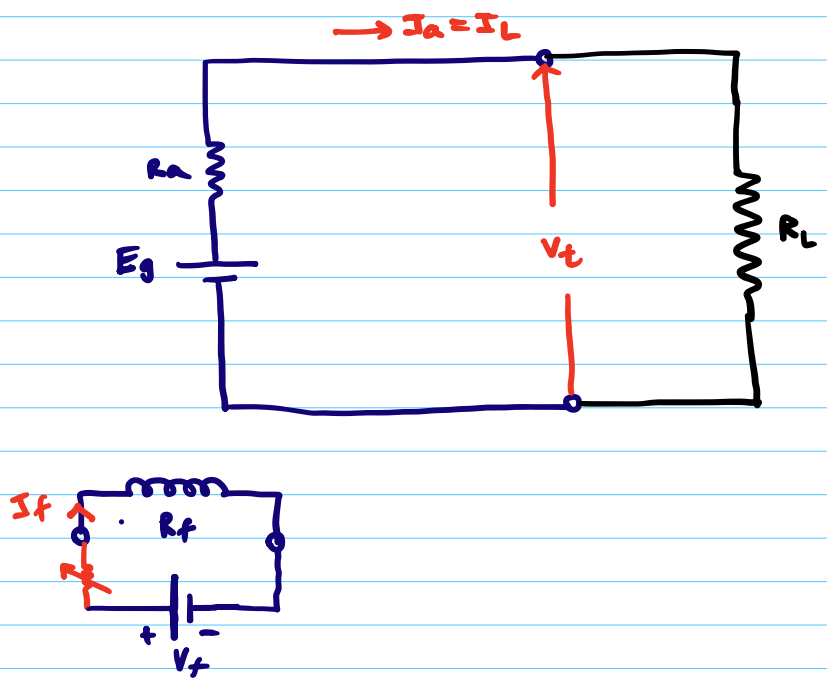
I_f (A)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0	1.2
E_g (V)	5	20	40	60	79	93	102	114	120	125

The machine parameters are $R_a = 0.2\,\Omega$ and $R_f = 100\,\Omega$. The machine is driven at 1200 rpm and the field current is adjusted at 0.8 A. A load resistance, $R_L = 2\,\Omega$ is connected to the armature terminals. Neglect armature reaction effect.

(a) Determine the quantity k_e for the machine.

(b) Determine the generated voltage and the load current.

(c) Calculate the electrical power supplied to the load.

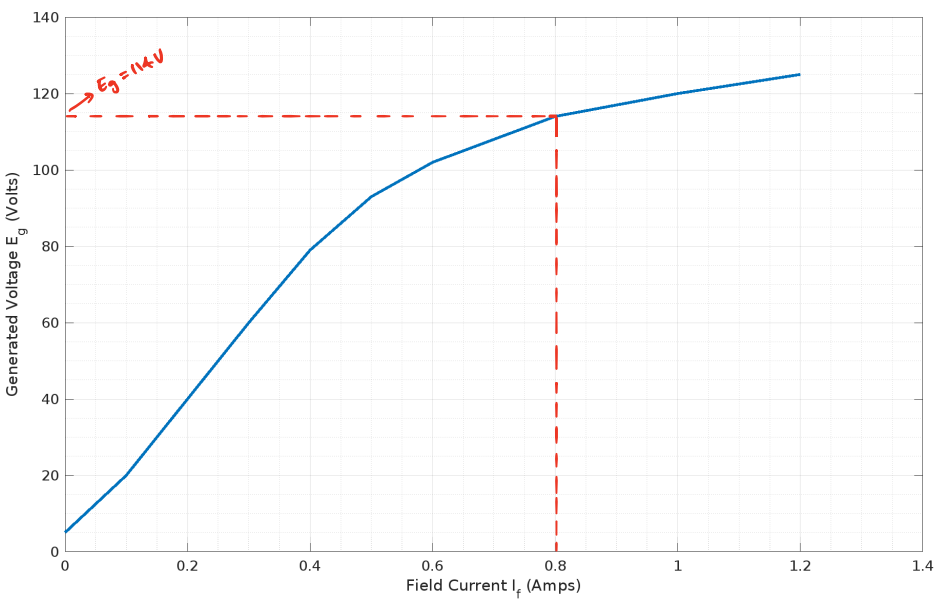


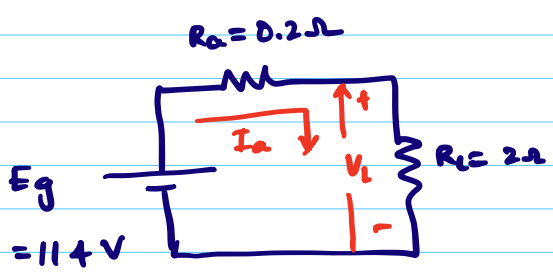
$V_t = 120V \qquad P_o = 6\,KW$

$$I_a = \frac{P_o}{V_t}$$

$$= 50\,A$$

$I_a = 50\,A$ (rated)





$$I_a = \frac{E_g}{R_a + R_L}$$

$$I_a = \frac{E_g}{R_a + R_L}$$

$$= \frac{114}{0.2 + 2}$$

$$I_a = 51.81818 \text{ A}$$

$$V_t = 114 \frac{2}{0.2 + 2}$$

$$V_t = 103.63636 \text{ V}$$

$$E_g = k_e \omega_m$$

$$k_e = \frac{E_g}{\omega_m}$$

$$= \frac{114}{\frac{2\pi}{60} \times 1200}$$

$$k_e = 0.907183 \text{ v/rad/sec}$$

Electrical output to load

$$P_{Load} = V_t I_a$$

$$= 103.63636 \times 51.8181$$

$$P_{Load} = 5370.2479 \text{ W}$$

(or) $P_{Load} = I_a^2 R_L$

$= (51.8181)^2 \times 2$

$P_{Load} = 5370.2479 \text{ W}$

